The Æther-flow Interpretation of Relativity

Primitive-Reservoir Observer-Reduction
Completion-Balance Ambient Dynamics
Necessity / Uniqueness from Combined Response Substrate Law

Abstract

The current benchmark-facing theorem on the active primitive-reservoir observer-reduction line already sharpened the burden below the exact-shell affine nucleus frontier once more: the minimal completion-state shell-trace core $\mathfrak{C}^{\mathrm{tr}}$ is now known to be a canonical and necessary reduction of the completion-balance ambient dynamics package $\mathfrak{A}^{\mathrm{bal}}$. The remaining honest task is therefore no longer to restate the trace core, the completion core, the exact-shell affine nucleus, or another ambient relay that preserves the same ambient package. It is to justify why the ambient package itself should hold.

The present note discharges that burden in a stronger compressed form than the preserved historical relays. For each sector it starts from a deeper combined response substrate law consisting of a compact convex primitive substrate body, one affine combined response map to observer data, ambient shell response, and hidden response, an exact factorization of the response image over the hidden-response body, a positively gapped hidden response operator, symmetry and time-reversal actions preserving that response image, an exact primitive-shell realization on the zero shell/hidden response locus, fixed-data shell solvability, and a response-kernel coercivity statement on the primitive shell.

From that single response law one derives canonically every constituent of the completion-balance ambient dynamics package: the ambient admissible body, the affine observer-data and shell-response readouts, the invariant affine zero-hidden slice, the ambient balance operator with kernel equal to the zero-hidden translation space, the exact body splitting over zero-hidden and hidden directions, the exact zero-response shell realization, fixed-data shell solvability on the zero-hidden slice, the inert completion kernel, and the completion-balance coercive inequality. Within this combined-response class the ambient package is not merely available. It is necessary and unique up to the obvious factor-preserving orthogonal identification on the hidden response factor.

The benchmark boundary remains fixed throughout. The one operative metric is still exactly $g^\sharp$, the ordinary matter route is unchanged, there is no second operative metric, the low-energy matter law is not enlarged, and stronger first-principles promotion language is not licensed. The gain is therefore a disciplined benchmark-facing compression: the current ambient package is no longer an independently unexplained stopping point but a canonical output of a deeper explicit substrate law.

1. Introduction

The active derivational program of Æther-flow Interpretation of Relativity has reached a more exact upstream burden than before. The primitive-observer-reduction datum theorem, the defect-fiber factorization theorem, the one-metric output theorem, the chain-forcing theorem, the selector

theorem, the stationary-construction theorem, the explicit-package theorem, the primitive-normal-form theorem, the ambient-shell-law theorem, the combined-response theorem, the completion-core theorem, the ambient trace-core forcing theorem, and the later ambient-dynamics routing note have already done the work that was honestly available at their respective layers. The latest of those results is especially important because it changed what now counts as front-line progress. Once the trace core $\mathfrak{C}^{\mathrm{tr}}$ is itself forced by the deeper ambient package $\mathfrak{A}^{\mathrm{bal}}$, another trace-core restatement or another ambient relay that preserves that same ambient package no longer counts as the honest next step.

The remaining honest burden is therefore deeper and more specific. The open task is now to show why the completion-balance ambient dynamics package itself should be forced on the active line rather than becoming the new disciplined stopping point. That is the burden addressed here.

The present note does not reopen the benchmark effective package. It does not alter the one operative completed metric, it does not introduce a second operative metric, and it does not enlarge the low-energy matter law. Its positive move is narrower and upstream. It shows that the current completion-balance ambient dynamics package is already a canonical and necessary reduction of a deeper combined response substrate law with hidden-sector gap.

Framework and claim boundary. Æther / Æther-flow framework denotes the broader Æther / Æther-flow framework built on the claims that the Æther is the underlying four-dimensional substrate of reality, the Æther-flow is its intrinsic ordered motion, observed three-dimensional space is the local experiential slice of that deeper substrate, S-time is the experienced order of change arising from matter, light, and the Æther-flow, and observed expansion is the three-dimensional appearance of deeper four-dimensional ordered motion. Within that framework, Æther-flow Interpretation of Relativity denotes the exact-closure theory in which the effective gravitational dynamics are adopted to be exactly Einsteinian with universal matter coupling. In the active sequence, *adoption* means use of that established relativistic dynamics without claiming substrate derivation, while *derivation* is reserved for a first-principles recovery from explicit substrate variables.

The present note stays strictly inside that boundary. It does not claim that final substrate microphysics has already been found. Its narrower theorem is only that, on the active primitive-observer-reduction line, the completion-balance ambient dynamics package $\mathfrak{A}^{\mathrm{bal}}$ is itself a forced reduction of a deeper combined response substrate law, and is unique there up to the obvious hidden-factor identification that preserves the recorded data and shell coordinates.

2. Fixed Primitive Continuation Data

Definition 1 (Recorded primitive continuation data)**.** *Fix the following continuation data from the active primitive-reservoir observer-reduction line.*

1. *For each sector label* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*, a primitive forcing shell*

$$\mathcal{I}_\sigma^{\mathrm{for}},$$

a visible reduction map

$$\mathcal{R}_\sigma : \mathcal{I}_\sigma^{\mathrm{for}} \to X_\sigma,$$

and shell-level current and equilibrium carriers

$$\mathfrak{C}_\sigma^{\mathrm{cur}} : \mathcal{I}_\sigma^{\mathrm{for}} \to Z_\sigma^{\mathrm{cur}}, \qquad \mathfrak{C}_\sigma^{\mathrm{eq}} : \mathcal{I}_\sigma^{\mathrm{for}} \to Z_\sigma^{\mathrm{eq}}.$$

2. *The product shell*

$$\mathcal{S}_{\mathrm{prim}} \equiv \mathcal{I}_{\mathrm{car}}^{\mathrm{for}} \times \mathcal{I}_{\mathrm{cur}}^{\mathrm{for}} \times \mathcal{I}_{\mathrm{hom}}^{\mathrm{for}}.$$

3. *Fixed shell extractors*

$$\mathscr{E}_\sigma^{\mathrm{cur}} : Z_\sigma^{\mathrm{cur}} \to Y_\sigma^{\mathrm{cur}}, \qquad \mathscr{E}_\sigma^{\mathrm{eq}} : Z_\sigma^{\mathrm{eq}} \to Y_\sigma^{\mathrm{eq}},$$

together with the recorded shell composites

$$\mathcal{J}_\sigma^{\mathrm{prim}} = \mathscr{E}_\sigma^{\mathrm{cur}} \circ \mathfrak{C}_\sigma^{\mathrm{cur}}, \qquad \Theta_\sigma^{\mathrm{prim}} = \mathscr{E}_\sigma^{\mathrm{eq}} \circ \mathfrak{C}_\sigma^{\mathrm{eq}}. \tag{1}$$

4. *The recorded metric and ordinary-matter outputs*

$$\mathscr{G}_{\mathrm{out}} : \mathcal{S}_{\mathrm{prim}} \to \{\textit{observer metrics}\}, \qquad \mathscr{T}_{\mathrm{out}} : \mathcal{S}_{\mathrm{prim}} \times \Psi \to \{\textit{observer stress tensors}\},$$

satisfying

$$\mathscr{G}_{\mathrm{out}}(\mathbf{w}) = g^\sharp, \qquad \mathscr{T}_{\mathrm{out}}(\mathbf{w}, \psi) = T_{\mu\nu}^{\mathrm{matter}}[g^\sharp, \psi] \tag{2}$$

for every $\mathbf{w} \in \mathcal{S}_{\mathrm{prim}}$ *and every matter configuration* $\psi \in \Psi$.

Remark 1. *Definition 1 is not reopened here. The primitive shell data and the benchmark one-metric / ordinary-matter outputs remain fixed continuation data. The new burden begins one layer deeper again: why the current ambient package* $\mathfrak{A}^{\mathrm{bal}}$ *that now carries the active line should itself be forced.*

3. Target Ambient Dynamics

Definition 2 (Completion-balance ambient dynamics with inert completion kernel)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$. *A* completion-balance ambient dynamics package *for sector* σ *consists of the following data.*

1. *A finite-dimensional Euclidean ambient state space*

$$\mathcal{Z}_\sigma,$$

the observer-data space

$$\mathcal{Y}_\sigma^{\mathrm{obs}} \equiv \widehat{X}_\sigma \times Z_\sigma^{\mathrm{cur}} \times Z_\sigma^{\mathrm{eq}},$$

an ambient shell-response space

$$E_\sigma^{\mathrm{amb}},$$

and a compact convex ambient admissible body

$$\mathcal{B}_\sigma^{\mathrm{bal}} \subset \mathcal{Z}_\sigma$$

with nonempty interior.

2. *An affine observer-data readout*

$$\Lambda_\sigma^{\mathrm{dat}} : \mathcal{Z}_\sigma \to \mathcal{Y}_\sigma^{\mathrm{obs}}$$

and an affine shell-response readout

$$\Lambda_\sigma^{\mathrm{sh}} : \mathcal{Z}_\sigma \to E_\sigma^{\mathrm{amb}}.$$

3. *An invariant affine zero-hidden slice*

$$\mathcal{Z}_\sigma^0 \subset \mathcal{Z}_\sigma$$

with translation space

$$V_\sigma^0 \equiv T_z \mathcal{Z}_\sigma^0 \qquad (z \in \mathcal{Z}_\sigma^0),$$

and hidden complement

$$H_\sigma \equiv V_\sigma^{0\perp} \subset \mathcal{Z}_\sigma.$$

4. *A positive self-adjoint ambient balance operator*

$$\mathcal{K}_\sigma^{\mathrm{bal}} : \mathcal{Z}_\sigma \to \mathcal{Z}_\sigma$$

whose kernel is exactly the zero-hidden translation space,

$$\ker \mathcal{K}_\sigma^{\mathrm{bal}} = V_\sigma^0,$$

and whose restriction to the hidden complement obeys

$$\left\langle \eta, \mathcal{K}_\sigma^{\mathrm{bal}} \eta \right\rangle \geq \kappa_\sigma^{\mathrm{hid}} \|\eta\|^2 \qquad \textit{for all } \eta \in H_\sigma \tag{3}$$

for some $\kappa_\sigma^{\mathrm{hid}} > 0$.

5. *The hidden projection body*

$$D_\sigma \equiv \left\{ \eta \in H_\sigma : z + \eta \in \mathcal{B}_\sigma^{\mathrm{bal}} \textit{ for some } z \in \mathcal{B}_\sigma^{\mathrm{bal}} \cap \mathcal{Z}_\sigma^0 \right\}$$

is compact and convex, satisfies

$$0 \in \operatorname{int} D_\sigma,$$

and the ambient admissible body splits exactly as

$$\mathcal{B}_\sigma^{\mathrm{bal}} = \left\{ z + \eta : z \in \mathcal{B}_\sigma^{\mathrm{bal}} \cap \mathcal{Z}_\sigma^0, \quad \eta \in D_\sigma \right\}. \tag{4}$$

6. *A compact symmetry group*

$$\mathbb{B}_\sigma \subset O(\mathcal{Z}_\sigma) \times O(\mathcal{Y}_\sigma^{\mathrm{obs}}) \times O(E_\sigma^{\mathrm{amb}})$$

and an orthogonal involution

$$\vartheta_\sigma^{\mathrm{bal}} \in O(\mathcal{Z}_\sigma) \times O(\mathcal{Y}_\sigma^{\mathrm{obs}}) \times O(E_\sigma^{\mathrm{amb}})$$

preserving $\mathcal{B}_\sigma^{\mathrm{bal}}$, $\mathcal{Z}_\sigma^0$, $\Lambda_\sigma^{\mathrm{dat}}$, $\Lambda_\sigma^{\mathrm{sh}}$, *and* $\mathcal{K}_\sigma^{\mathrm{bal}}$.

7. *A smooth embedding*

$$\jmath_\sigma : X_\sigma \hookrightarrow \widehat{X}_\sigma$$

and a smooth zero-response shell realization

$$\zeta_\sigma^{\mathrm{bal}} : \mathcal{I}_\sigma^{\mathrm{for}} \hookrightarrow \mathcal{Z}_\sigma$$

whose image is exactly

$$\mathcal{B}_\sigma^{\mathrm{bal}} \cap \mathcal{Z}_\sigma^0 \cap (\Lambda_\sigma^{\mathrm{sh}})^{-1}(0),$$

and such that

$$\Lambda_\sigma^{\mathrm{dat}}(\zeta_\sigma^{\mathrm{bal}}(w)) = (\jmath_\sigma(\mathcal{R}_\sigma(w)), \mathfrak{C}_\sigma^{\mathrm{cur}}(w), \mathfrak{C}_\sigma^{\mathrm{eq}}(w)) \tag{5}$$

for every $w \in \mathcal{I}_\sigma^{\mathrm{for}}$.

8. *Fixed-data shell solvability on the zero-hidden slice:*

$$\Lambda_\sigma^{\mathrm{dat}}(\mathcal{B}_\sigma^{\mathrm{bal}} \cap \mathcal{Z}_\sigma^0) = \Lambda_\sigma^{\mathrm{dat}}\Big(\mathcal{B}_\sigma^{\mathrm{bal}} \cap \mathcal{Z}_\sigma^0 \cap (\Lambda_\sigma^{\mathrm{sh}})^{-1}(0)\Big). \tag{6}$$

9. *The inert completion kernel*

$$K_\sigma^0 \equiv \ker(D\Lambda_\sigma^{\mathrm{dat}}|_{V_\sigma^0}) \cap \ker(D\Lambda_\sigma^{\mathrm{sh}}|_{V_\sigma^0})$$

and the orthogonal completion-state projector

$$P_\sigma^{\mathrm{co}} : V_\sigma^0 \to K_\sigma^{0\perp} \cap V_\sigma^0.$$

10. *Completion-balance coercivity on data-invisible zero-hidden directions: there exists* $\kappa_\sigma^{\mathrm{co}} > 0$ *such that for every*

$$w \in \mathcal{I}_\sigma^{\mathrm{for}}, \qquad \delta z \in T_{\zeta_\sigma^{\mathrm{bal}}(w)}\mathcal{Z}_\sigma^0 = V_\sigma^0$$

satisfying

$$D\Lambda_\sigma^{\mathrm{dat}}\,\delta z = 0,$$

one has

$$\|D\Lambda_\sigma^{\mathrm{sh}}\,\delta z\|^2 \geq \kappa_\sigma^{\mathrm{co}}\,\|P_\sigma^{\mathrm{co}}\delta z\|^2. \tag{7}$$

Remark 2. *Definition 2 is the target package whose derivational status is now at issue. The purpose here is not to reintroduce it as another deeper stopping point, but to show that it is already a canonical output of a deeper combined response substrate law.*

4. Combined Response Substrate Law

Definition 3 (Combined response substrate law with hidden-sector gap)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$. *A* combined response substrate law with hidden-sector gap *for sector* σ *consists of the following data.*

1. *A finite-dimensional Euclidean primitive substrate space*

$$\mathcal{M}_\sigma$$

with a compact convex body

$$\mathcal{P}_\sigma \subset \mathcal{M}_\sigma.$$

2. *An observer-data target space*

$$\mathcal{Y}_\sigma^{\mathrm{obs}} \equiv \widehat{X}_\sigma \times Z_\sigma^{\mathrm{cur}} \times Z_\sigma^{\mathrm{eq}},$$

an ambient shell target

$$E_\sigma^{\mathrm{amb}},$$

a hidden response space

$$H_\sigma^{\mathrm{resp}},$$

and the product response space

$$\mathcal{R}_\sigma \equiv \mathcal{Y}_\sigma^{\mathrm{obs}} \oplus E_\sigma^{\mathrm{amb}} \oplus H_\sigma^{\mathrm{resp}}.$$

3. *An affine combined response map*

$$\Xi_\sigma = (\Xi_\sigma^{\mathrm{dat}}, \Xi_\sigma^{\mathrm{sh}}, \Xi_\sigma^{\mathrm{hid}}) : \mathcal{M}_\sigma \to \mathcal{R}_\sigma.$$

4. *The response image body*

$$\mathcal{B}_\sigma^{\mathrm{resp}} \equiv \Xi_\sigma(\mathcal{P}_\sigma) \tag{8}$$

is compact and convex, and its hidden-zero slice

$$\mathcal{B}_\sigma^{\mathrm{resp}} \cap \left(\mathcal{Y}_\sigma^{\mathrm{obs}} \oplus E_\sigma^{\mathrm{amb}} \oplus \{0\}\right)$$

has nonempty relative interior in

$$\mathcal{Y}_\sigma^{\mathrm{obs}} \oplus E_\sigma^{\mathrm{amb}} \oplus \{0\}.$$

5. *A compact convex hidden-response body*

$$D_\sigma^{\mathrm{resp}} \subset H_\sigma^{\mathrm{resp}}$$

with

$$0 \in \operatorname{int} D_\sigma^{\mathrm{resp}},$$

such that the response image factorizes exactly as

$$\mathcal{B}_\sigma^{\mathrm{resp}} = \left\{(u,e,n) : (u,e,0) \in \mathcal{B}_\sigma^{\mathrm{resp}}, \quad n \in D_\sigma^{\mathrm{resp}}\right\}. \tag{9}$$

Equivalently, the map

$$(u,e,n) \longmapsto \big((u,e,0),n\big) \tag{10}$$

is a bijection from $\mathcal{B}_\sigma^{\mathrm{resp}}$ *onto*

$$\Big(\mathcal{B}_\sigma^{\mathrm{resp}} \cap \left(\mathcal{Y}_\sigma^{\mathrm{obs}} \oplus E_\sigma^{\mathrm{amb}} \oplus \{0\}\right)\Big) \times D_\sigma^{\mathrm{resp}}.$$

6. *A positive self-adjoint hidden-response operator*

$$K_\sigma^{\mathrm{resp}} : H_\sigma^{\mathrm{resp}} \to H_\sigma^{\mathrm{resp}}$$

with lower bound

$$\langle \eta, K_\sigma^{\mathrm{resp}} \eta \rangle \geq \kappa_\sigma^{\mathrm{hid}} \, \|\eta\|^2 \qquad \textit{for all } \eta \in H_\sigma^{\mathrm{resp}} \tag{11}$$

for some $\kappa_\sigma^{\mathrm{hid}} > 0$.

7. *A compact symmetry group*

$$\mathbb{H}_\sigma \subset O(\mathcal{Y}_\sigma^{\mathrm{obs}}) \times O(E_\sigma^{\mathrm{amb}}) \times O(H_\sigma^{\mathrm{resp}})$$

and an orthogonal involution

$$\rho_\sigma \in O(\mathcal{Y}_\sigma^{\mathrm{obs}}) \times O(E_\sigma^{\mathrm{amb}}) \times O(H_\sigma^{\mathrm{resp}})$$

preserving $\mathcal{B}_\sigma^{\mathrm{resp}}$*, the hidden-zero slice*

$$\mathcal{Y}_\sigma^{\mathrm{obs}} \oplus E_\sigma^{\mathrm{amb}} \oplus \{0\},$$

the zero shell value $0 \in E_\sigma^{\mathrm{amb}}$*, and* K_σ^{resp}.

8. *A smooth embedding*

$$\jmath_\sigma : X_\sigma \hookrightarrow \widehat{X}_\sigma$$

and a smooth shell realization map

$$\zeta_\sigma : \mathcal{I}_\sigma^{\mathrm{for}} \hookrightarrow \mathcal{M}_\sigma$$

such that

$$\zeta_\sigma(\mathcal{I}_\sigma^{\mathrm{for}}) \subset \mathcal{P}_\sigma \cap (\Xi_\sigma^{\mathrm{sh}})^{-1}(0) \cap (\Xi_\sigma^{\mathrm{hid}})^{-1}(0),$$

the composition

$$\zeta_\sigma^{\mathrm{bal}} \equiv \Xi_\sigma \circ \zeta_\sigma \tag{12}$$

is a diffeomorphism onto

$$\mathcal{B}_\sigma^{\mathrm{resp}} \cap (\mathcal{Y}_\sigma^{\mathrm{obs}} \oplus \{0\} \oplus \{0\}),$$

and

$$\Xi_\sigma^{\mathrm{dat}}(\zeta_\sigma(w)) = (\jmath_\sigma(\mathcal{R}_\sigma(w)), \mathfrak{C}_\sigma^{\mathrm{cur}}(w), \mathfrak{C}_\sigma^{\mathrm{eq}}(w)) \tag{13}$$

for every $w \in \mathcal{I}_\sigma^{\mathrm{for}}$.

9. *Zero-response shell solvability on fixed data fibers:*

$$\mathrm{pr}_{\mathcal{Y}_\sigma^{\mathrm{obs}}}(\mathcal{B}_\sigma^{\mathrm{resp}}) = \mathrm{pr}_{\mathcal{Y}_\sigma^{\mathrm{obs}}}\Big(\mathcal{B}_\sigma^{\mathrm{resp}} \cap (\mathcal{Y}_\sigma^{\mathrm{obs}} \oplus \{0\} \oplus \{0\})\Big). \tag{14}$$

10. *Response-kernel coercivity on the primitive shell: there exists* $\kappa_\sigma^{\mathrm{resp}} > 0$ *such that for every*

$$w \in \mathcal{I}_\sigma^{\mathrm{for}}, \qquad \delta m \in T_{\zeta_\sigma(w)}\mathcal{M}_\sigma$$

satisfying

$$D\Xi_\sigma^{\mathrm{dat}}(\zeta_\sigma(w))\,\delta m = 0,$$

one has

$$\left\|D\Xi_\sigma^{\mathrm{sh}}(\zeta_\sigma(w))\,\delta m\right\|^2 + \left\langle D\Xi_\sigma^{\mathrm{hid}}(\zeta_\sigma(w))\,\delta m, K_\sigma^{\mathrm{resp}} D\Xi_\sigma^{\mathrm{hid}}(\zeta_\sigma(w))\,\delta m\right\rangle \geq \kappa_\sigma^{\mathrm{resp}}\,\|\delta m\|^2. \tag{15}$$

Remark 3. *Definition 3 is deeper than Definition 2 in exactly the way now needed. It does not stipulate separately an ambient admissible body, a zero-hidden slice, an ambient balance operator, and a completion-balance coercive package. It starts from one compact convex primitive substrate body and one affine combined response map whose factorized image already contains those ambient constituents in compressed form.*

5. Canonical Ambient Dynamics

Definition 4 (Canonical ambient dynamics induced by the combined response)**.** *Assume Definition 3. Define the canonical completion-balance ambient dynamics package by:*

1. *The ambient state space is the response space:*

$$\mathcal{Z}_\sigma \equiv \mathcal{R}_\sigma = \mathcal{Y}_\sigma^{\mathrm{obs}} \oplus E_\sigma^{\mathrm{amb}} \oplus H_\sigma^{\mathrm{resp}}.$$

2. *The ambient admissible body is the response image body:*

$$\mathcal{B}_\sigma^{\mathrm{bal}} \equiv \mathcal{B}_\sigma^{\mathrm{resp}}.$$

3. *The observer-data and shell-response readouts are the coordinate projections*

$$\Lambda_\sigma^{\mathrm{dat}}(u,e,n) \equiv u, \qquad \Lambda_\sigma^{\mathrm{sh}}(u,e,n) \equiv e. \tag{16}$$

4. *The zero-hidden slice is the hidden-zero response slice*

$$\mathcal{Z}_\sigma^0 \equiv \mathcal{Y}_\sigma^{\mathrm{obs}} \oplus E_\sigma^{\mathrm{amb}} \oplus \{0\}, \qquad V_\sigma^0 \equiv \mathcal{Y}_\sigma^{\mathrm{obs}} \oplus E_\sigma^{\mathrm{amb}} \oplus \{0\}, \tag{17}$$

   and the hidden complement is

$$H_\sigma \equiv \{0\} \oplus \{0\} \oplus H_\sigma^{\mathrm{resp}}. \tag{18}$$

5. *The ambient balance operator is*

$$\mathcal{K}_\sigma^{\mathrm{bal}}(u,e,n) \equiv (0,0,K_\sigma^{\mathrm{resp}} n). \tag{19}$$

6. *The hidden projection body is*

$$D_\sigma \equiv \{(0,0,n) : n \in D_\sigma^{\mathrm{resp}}\}. \tag{20}$$

7. The symmetry group and time reversal are

$$\mathbb{B}_\sigma \equiv \mathbb{H}_\sigma, \qquad \vartheta_\sigma^{\mathrm{bal}} \equiv \rho_\sigma.$$

8. The zero-response shell realization is exactly (12)*.*

Proposition 1 (The combined response law induces completion-balance ambient dynamics)**.** *Assume Definitions 3 and 4. Then the canonical data of Definition 4 satisfy every requirement of Definition 2.*

Proof. Because Ξ_σ is affine and $\mathcal{P}_\sigma$ is compact and convex, $\mathcal{B}_\sigma^{\mathrm{bal}} = \mathcal{B}_\sigma^{\mathrm{resp}}$ is compact and convex. Equation (16) defines affine readouts on the finite-dimensional Euclidean product space $\mathcal{Z}_\sigma$.

Equation (17) defines an invariant affine zero-hidden slice, and because the ambient norm is the product Euclidean norm, its orthogonal complement is exactly H_σ from (18). The operator (19) is self-adjoint and positive semidefinite. Its kernel is exactly

$$\ker \mathcal{K}_\sigma^{\mathrm{bal}} = \mathcal{Y}_\sigma^{\mathrm{obs}} \oplus E_\sigma^{\mathrm{amb}} \oplus \{0\} = V_\sigma^0,$$

because K_σ^{resp} is positive definite on H_σ^{resp}. For

$$\eta = (0, 0, n) \in H_\sigma,$$

equation (11) gives

$$\left\langle \eta, \mathcal{K}_\sigma^{\mathrm{bal}} \eta \right\rangle = \langle n, K_\sigma^{\mathrm{resp}} n \rangle \geq \kappa_\sigma^{\mathrm{hid}} \|n\|^2 = \kappa_\sigma^{\mathrm{hid}} \|\eta\|^2,$$

which is exactly (3).

The hidden projection body of Definition 2 is exactly D_σ from (20), and the exact body splitting (4) is precisely the factorization (9): an element (u, e, n) lies in $\mathcal{B}_\sigma^{\mathrm{bal}}$ if and only if $(u, e, 0) \in \mathcal{B}_\sigma^{\mathrm{bal}} \cap \mathcal{Z}_\sigma^0$ and $(0, 0, n) \in D_\sigma$. Because $0 \in \operatorname{int} D_\sigma^{\mathrm{resp}}$, we also have $0 \in \operatorname{int} D_\sigma$.

The symmetry and time-reversal requirements follow immediately because every element of $\mathbb{H}_\sigma$ and ρ_σ preserves $\mathcal{B}_\sigma^{\mathrm{resp}}$, the hidden-zero response slice, the zero shell value, and K_σ^{resp}, hence preserves $\mathcal{B}_\sigma^{\mathrm{bal}}$, $\mathcal{Z}_\sigma^0$, $\Lambda_\sigma^{\mathrm{dat}}$, $\Lambda_\sigma^{\mathrm{sh}}$, and $\mathcal{K}_\sigma^{\mathrm{bal}}$.

For the shell realization,

$$\zeta_\sigma^{\mathrm{bal}}(\mathcal{I}_\sigma^{\mathrm{for}}) = \mathcal{B}_\sigma^{\mathrm{resp}} \cap (\mathcal{Y}_\sigma^{\mathrm{obs}} \oplus \{0\} \oplus \{0\})$$

by Definition 3. But this set is exactly

$$\mathcal{B}_\sigma^{\mathrm{bal}} \cap \mathcal{Z}_\sigma^0 \cap (\Lambda_\sigma^{\mathrm{sh}})^{-1}(0),$$

and (13) becomes (5).

For fixed-data shell solvability, the factorization (9) implies

$$\Lambda_\sigma^{\mathrm{dat}}(\mathcal{B}_\sigma^{\mathrm{bal}} \cap \mathcal{Z}_\sigma^0) = \mathrm{pr}_{\mathcal{Y}_\sigma^{\mathrm{obs}}}(\mathcal{B}_\sigma^{\mathrm{resp}}),$$

because every point of $\mathcal{B}_\sigma^{\mathrm{resp}}$ has the same data coordinate as a hidden-zero point of $\mathcal{B}_\sigma^{\mathrm{resp}}$. Equation (14) then gives

$$\mathrm{pr}_{\mathcal{Y}_\sigma^{\mathrm{obs}}}(\mathcal{B}_\sigma^{\mathrm{resp}}) = \mathrm{pr}_{\mathcal{Y}_\sigma^{\mathrm{obs}}}\Big(\mathcal{B}_\sigma^{\mathrm{resp}} \cap (\mathcal{Y}_\sigma^{\mathrm{obs}} \oplus \{0\} \oplus \{0\})\Big),$$

which is exactly (6).

For the inert completion kernel, note that on the zero-hidden slice the derivatives of $\Lambda_\sigma^{\mathrm{dat}}$ and $\Lambda_\sigma^{\mathrm{sh}}$ are simply the coordinate projections

$$D\Lambda_\sigma^{\mathrm{dat}}(u,e,0) = u, \qquad D\Lambda_\sigma^{\mathrm{sh}}(u,e,0) = e.$$

Hence

$$K_\sigma^0 = \ker(D\Lambda_\sigma^{\mathrm{dat}}|_{V_\sigma^0}) \cap \ker(D\Lambda_\sigma^{\mathrm{sh}}|_{V_\sigma^0}) = \{0\},$$

so P_σ^{co} is the identity on V_σ^0.

Finally, let

$$w \in \mathcal{I}_\sigma^{\mathrm{for}}, \qquad \delta z \in T_{\zeta_\sigma^{\mathrm{bal}}(w)}\mathcal{Z}_\sigma^0 = V_\sigma^0$$

satisfy

$$D\Lambda_\sigma^{\mathrm{dat}}\,\delta z = 0.$$

Writing $\delta z = (\delta u, \delta e, 0)$, the data-kernel condition gives $\delta u = 0$. Therefore

$$\|D\Lambda_\sigma^{\mathrm{sh}}\,\delta z\|^2 = \|\delta e\|^2 = \|\delta z\|^2 = \|P_\sigma^{\mathrm{co}}\delta z\|^2.$$

Thus (7) holds with $\kappa_\sigma^{\mathrm{co}} = 1$. All requirements of Definition 2 are satisfied. □

6. Forced Ambient Constituents and Uniqueness

Proposition 2 (All ambient constituents are forced by the combined response law)**.** *Assume Definition 3. Then the following ambient-level constituents are fixed by the combined response law itself.*

1. *The ambient admissible body is necessarily the response image body $\mathcal{B}_\sigma^{\mathrm{bal}} = \mathcal{B}_\sigma^{\mathrm{resp}}$.*
2. *The affine observer-data and shell-response readouts are necessarily the coordinate projections $\Lambda_\sigma^{\mathrm{dat}}(u,e,n) = u$ and $\Lambda_\sigma^{\mathrm{sh}}(u,e,n) = e$.*
3. *The invariant zero-hidden slice is necessarily the hidden-zero response slice $\mathcal{Z}_\sigma^0 = \mathcal{Y}_\sigma^{\mathrm{obs}} \oplus E_\sigma^{\mathrm{amb}} \oplus \{0\}$, with hidden complement $H_\sigma = \{0\} \oplus \{0\} \oplus H_\sigma^{\mathrm{resp}}$.*
4. *The ambient balance operator is necessarily the hidden-response gap operator acting on the hidden factor as in* (19)*; its kernel is therefore exactly the zero-hidden translation space.*
5. *The hidden projection body, the exact body splitting, the symmetry / time-reversal structure, the exact zero-response shell realization, fixed-data shell solvability on the zero-hidden slice, the inert completion kernel $K_\sigma^0 = \{0\}$, and the completion-balance coercive inequality are all induced by the same combined response law and are not additional ambient freedom.*

Proof. Item (1) is immediate from Definition 4: once the combined response map and primitive substrate body are fixed, the only compatible ambient admissible body is their response image. Item (2) follows because the combined response law already separates the observer-data and shell-response coordinates in the response-space product; any ambient package compatible with that same law must use those same coordinate projections.

Item (3) follows because the only natural candidate for the zero-hidden locus singled out by the combined response law is the hidden-zero response slice itself, and the product Euclidean structure forces its orthogonal complement to be the embedded hidden response factor. Item (4) then follows because the ambient balance operator must vanish precisely on that hidden-zero slice and reproduce the given positive hidden quadratic law on the hidden factor, which is exactly (19).

Item (5) is the content of Proposition 1. Once the response image factorization, response symmetries, exact shell realization, fixed-data solvability, and hidden-response gap operator are fixed, all later ambient constituents descend directly from the same combined-response law and carry no additional independent freedom. □

Theorem 1 (Completion-balance ambient dynamics necessity / uniqueness from combined response law)**.** *Assume Definition 3. Then:*

1. *the canonical construction of Definition 4 yields a completion-balance ambient dynamics package in the sense of Definition 2;*
2. *every ambient-level object singled out by the current ambient frontier is forced by the combined response law: the compact convex ambient admissible body, the affine observer-data and shell-response readouts, the invariant affine zero-hidden slice, the ambient balance operator with kernel equal to the zero-hidden translation space, the hidden projection body and exact body splitting, the symmetry and time-reversal structure, the exact zero-response shell realization, fixed-data shell solvability on the zero-hidden slice, the inert completion kernel, and the completion-balance coercive inequality;*
3. *any other ambient package compatible with the same combined response law differs from the canonical one only by the obvious factor-preserving orthogonal identification on the hidden response factor that leaves the data and shell coordinates fixed;*
4. *consequently the completion-balance ambient dynamics layer is not an independent remaining freedom once the combined response law is fixed.*

Proof. Item (1) is Proposition 1. Item (2) is Proposition 2. For item (3), any compatible ambient package must preserve the response-space product decomposition

$$\mathcal{R}_\sigma = \mathcal{Y}_\sigma^{\mathrm{obs}} \oplus E_\sigma^{\mathrm{amb}} \oplus H_\sigma^{\mathrm{resp}},$$

must preserve the response image body, must keep the ambient data and shell maps equal to the first two coordinate projections, and must preserve the hidden quadratic law that yields (19). Therefore no freedom remains in the data or shell coordinates, and the only harmless reparameterization is the obvious orthogonal identification of the hidden response factor that preserves that hidden quadratic law. Item (4) is the routing consequence of items (1)–(3). □

7. Benchmark Preservation

Theorem 2 (The derived ambient package preserves the same one-metric benchmark package)**.** *Assume Definition 3. Then the benchmark output statements of Definition 1 remain unchanged:*

$$\mathscr{G}_{\mathrm{out}}(\mathbf{w}) = g^\sharp, \qquad \mathscr{T}_{\mathrm{out}}(\mathbf{w}, \psi) = T_{\mu\nu}^{\mathrm{matter}}[g^\sharp, \psi]$$

for every $\mathbf{w} \in \mathcal{S}_{\mathrm{prim}}$ *and every matter configuration* $\psi \in \Psi$*. Consequently the deeper combined response law and the induced ambient package introduce neither a second operative metric nor an enlarged low-energy matter law.*

Proof. The present note changes only the upstream origin of the current ambient package. It does not alter the primitive shell $\mathcal{S}_{\mathrm{prim}}$, the recorded metric map $\mathscr{G}_{\mathrm{out}}$, or the recorded matter map $\mathscr{T}_{\mathrm{out}}$. Those remain the fixed continuation data of Definition 1, so (2) continues to hold exactly. □

8. Main Theorems

Theorem 3 (The remaining burden moves below the ambient package)**.** *Assume the fixed primitive continuation data of Definition 1 and, for each sector* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*, a combined response substrate law in the sense of Definition 3. Then:*

1. *the completion-balance ambient dynamics package of Definition 2 is canonically induced by the combined response law;*
2. *the ambient admissible body, the affine observer-data and shell-response readouts, the invariant affine zero-hidden slice, the ambient balance operator with kernel equal to the zero-hidden translation space, the hidden projection body and exact body splitting, the symmetry and time-reversal structure, the exact zero-response shell realization, fixed-data shell solvability on the zero-hidden slice, the inert completion kernel, and the completion-balance coercive inequality are all forced by that same combined response law;*
3. *the one operative metric remains exactly* $g^{\sharp}$*, the ordinary matter route remains exactly the standard one through that metric, there is no second operative metric, and the low-energy matter law is not enlarged;*
4. *consequently the benchmark-facing burden after the ambient trace-core theorem and the ambient-dynamics routing criterion is discharged in the stronger form now available on the active line: the current ambient package* $\mathfrak{A}^{\mathrm{bal}}$ *is no longer an independently unexplained stopping point, but a necessary reduction of a deeper explicit combined response substrate law;*
5. *the next honest benchmark-facing attempt should therefore go one level below the current combined-response frontier: first try to derive or uniquely force that combined response law from still more primitive explicit substrate structure in a way that actually changes benchmark-facing status; if that cannot be done cleanly, the admissible fallback is to prove a genuinely smaller theorem-level collapse, sharper necessity result, or sharper uniqueness result strictly inside that combined response law while preserving the same benchmark one-metric package and claim boundary.*

Proof. Item (1) is Theorem 1(1). Item (2) is Theorem 1(2). Item (3) is Theorem 2. Items (4) and (5) are the combined routing consequence. □

Corollary 1 (What remains open after this theorem)**.** *After Theorem 3, the remaining honest burden is no longer to justify the completion-balance ambient dynamics package itself on the active primitive-observer-reduction line. The live burden is now one level below the current combined-response frontier. The next honest attempt should therefore first try to derive or uniquely force the combined response substrate law itself from still more primitive explicit substrate structure in a*

way that actually changes benchmark-facing status while preserving the same benchmark one-metric package and claim boundary. If that cannot be done cleanly, the admissible fallback is to prove a genuinely smaller theorem-level collapse, sharper necessity result, or sharper uniqueness result strictly inside that combined response law.

Proof. Theorem 3 converts the ambient package from the current live benchmark-facing burden into a canonical reduction of deeper combined response substrate law. What remains open is why that combined response law itself should hold, rather than becoming the new disciplined stopping point. Since another ambient-package restatement, another trace-core restatement, another completion-core restatement, another exact-shell affine nucleus restatement, or another same combined-response relay that preserves that same combined response law without shrinking or forcing it further would leave that burden unchanged, the next honest attempt must target the combined response law itself unless a sharper internal collapse, necessity result, or uniqueness result is the only clean benchmark-facing gain available. □

9. Conclusion

The positive result of this note is a direct forcing theorem for the current ambient stopping point. The previous ambient trace-core theorem already showed that the completion-state shell-trace core $\mathfrak{C}^{\mathrm{tr}}$ is a canonical reduction of the deeper completion-balance ambient dynamics package $\mathfrak{A}^{\mathrm{bal}}$. The present note then spends the first honest move available there: it shows that this ambient package is itself canonically induced by a deeper combined response substrate law.

That gain directly addresses the precise constituents that had become the live ambient burden: the ambient admissible body, the affine observer-data and shell-response readouts, the invariant zero-hidden slice, the ambient balance operator, the hidden body splitting, the exact zero-response shell realization, fixed-data shell solvability, the inert completion kernel, and the completion-balance coercive control. Within the recorded combined-response class those are no longer optional inserted ingredients. They are forced outputs, unique up to the harmless factor-preserving orthogonal identification on the hidden response factor.

The benchmark boundary remains fixed throughout. The same primitive shell remains the shell carrying the observer outputs, so the one operative metric is still exactly $g^{\sharp}$, the ordinary matter route is unchanged, there is no second operative metric, and the low-energy matter law is not enlarged. This is therefore a disciplined upstream derivational gain, not a final first-principles completion claim. The note does not yet derive the combined response law from ultimate substrate microphysics. The live burden is now one level below the current combined-response frontier: first try to derive or uniquely force that combined response law from still more primitive explicit substrate structure in a way that actually changes benchmark-facing status. If that cannot be done cleanly, the fallback is a genuinely smaller theorem-level collapse, sharper necessity result, or sharper uniqueness result strictly inside that combined response law. Another ambient-package restatement, trace-core restatement, completion-core restatement, exact-shell affine nucleus restatement, or another same combined-response relay that preserves that law without shrinking or forcing it further would not meet that burden, and no stronger first-principles promotion language is licensed yet.